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Constraining shallow slip deficit with phase gradient data

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SUMMARY

Coseismic slip models for large ($M_w > 7$) strike-slip earthquakes present a variety of shallow slip deficit (SSD). Accurate estimate of SSD is difficult, and it has been suggested that SSD are to some degree associated with fault-zone characteristics, incompleteness of data coverage as well as simplified model assumptions. Furthermore, SSD can also be sensitive to the amount of model smoothness adopted. Since phase gradient are sensitive to the missing shallow slip from our simulated data, we performed a synthetic test and presented a case study of the 2019 Ridgecrest earthquake sequence to validate that phase gradient from radar interferometry could help reveal the actual SSD for kinematic slip models even without enough near-fault observation. Our results indicate that even in the presence of a greater degree of observational gaps, the phase gradient can still nearly substitute for near-fault observations in constraining the shallow slip. Lastly, we provide a preferred coseismic slip model constrained by all available observations including phase gradient, but with 4-km data gap near the fault trace. This model results in ∼35 per cent SSD for the Ridgecrest earthquakes, matching previous estimates that incorporate near-field data. Considering the phase gradient approach is a straightforward mathematical operation, this approach may be applicable to other types of earthquakes. Notably, due to the smaller amplitude and lower signal-to-noise ratio for the phase gradient data, one needs to carefully balance the trade-offs among weights of different data sets and model smoothness.

Key words: Radar interferometry Reference; Satellite geodesy; Seismic cycle.

1 INTRODUCTION

Kinematic coseismic slip models inferred by space geodetic observations (e.g. interferometric synthetic aperture radar, InSAR and global navigation satellite systems, GNSS) indicate a slip reduction at shallow depths relative to deeper peak slips (e.g. 4–6 km) for major strike-slip earthquakes ($M_w > 7$), such as the 1992 M_w 7.3 Landers earthquake, the 1999 M_w 7.1 Hector Mine earthquake, the 1999 M_w 7.6 Izmit earthquake, the 2010 M_w 7.2 El Mayor–Cucapah earthquake, etc. (Simons *et al.* 2002; Fialko *et al.* 2004; Finzi *et al.* 2009; Wei *et al.* 2011). Such slip reduction from depth to surface has been termed as shallow slip deficit (SSD). Part of the coseismic slip deficit could be attributed to velocity-strengthening behaviour of uppermost brittle layer of Earth's shallowest crust (Marone *et al.* 1991; Scholz 1998) and could be compensated by shallow afterslip or interseismic creep over the long-term earthquake cycles (Reid 1910). Previous studies revealed that the appearance of a SSD is somewhat related to one or several of the following factors: (1) Fault-zone inelastic behaviours and off-fault distributed deformation may induce up to 25 per cent SSD if the inversion assumes a purely elastic Earth model (Kaneko & Fialko 2011; Roten *et al.* 2017); (2) The lack of near-fault geodetic observations due to strong decorrelation may generate an artificial SSD (Xu *et al.* 2016); (3) Correlated data noise, simplified elastic structure or topographic effect can also produce artificial SSD (Marchandon *et al.* 2021); (4) The different-scale patch size, influencing model roughness, could also affect the SSD (Huang *et al.* 2017; Fang *et al.* 2022); (5) Non-physical near-fault observations that conflict with fault model geometry may induce additional SSD artifacts to the slip model (Wang *et al.* 2025). Notably, model smoothness also affects SSD to a large degree though there are setups that can help explore the upper limit of missing shallow slip by selecting a rougher model setup (Xu *et al.* 2016). Thus, though the magnitude of SSD is critical for understanding fault kinematic and dynamic behaviours and the associated seismic hazards, the recovery of true SSD is very difficult.

The differences in simulated surface displacement profiles between a slip model with shallow deficit and one without deficit are only differentiable within a very close distance from the rupture (as noted in Simons *et al.* 2002) (Fig. 1). Such narrow region around the rupture usually lacks high-quality geodetic measurements due to shattered surfaces. In this study, we present that phase gradient (i.e.

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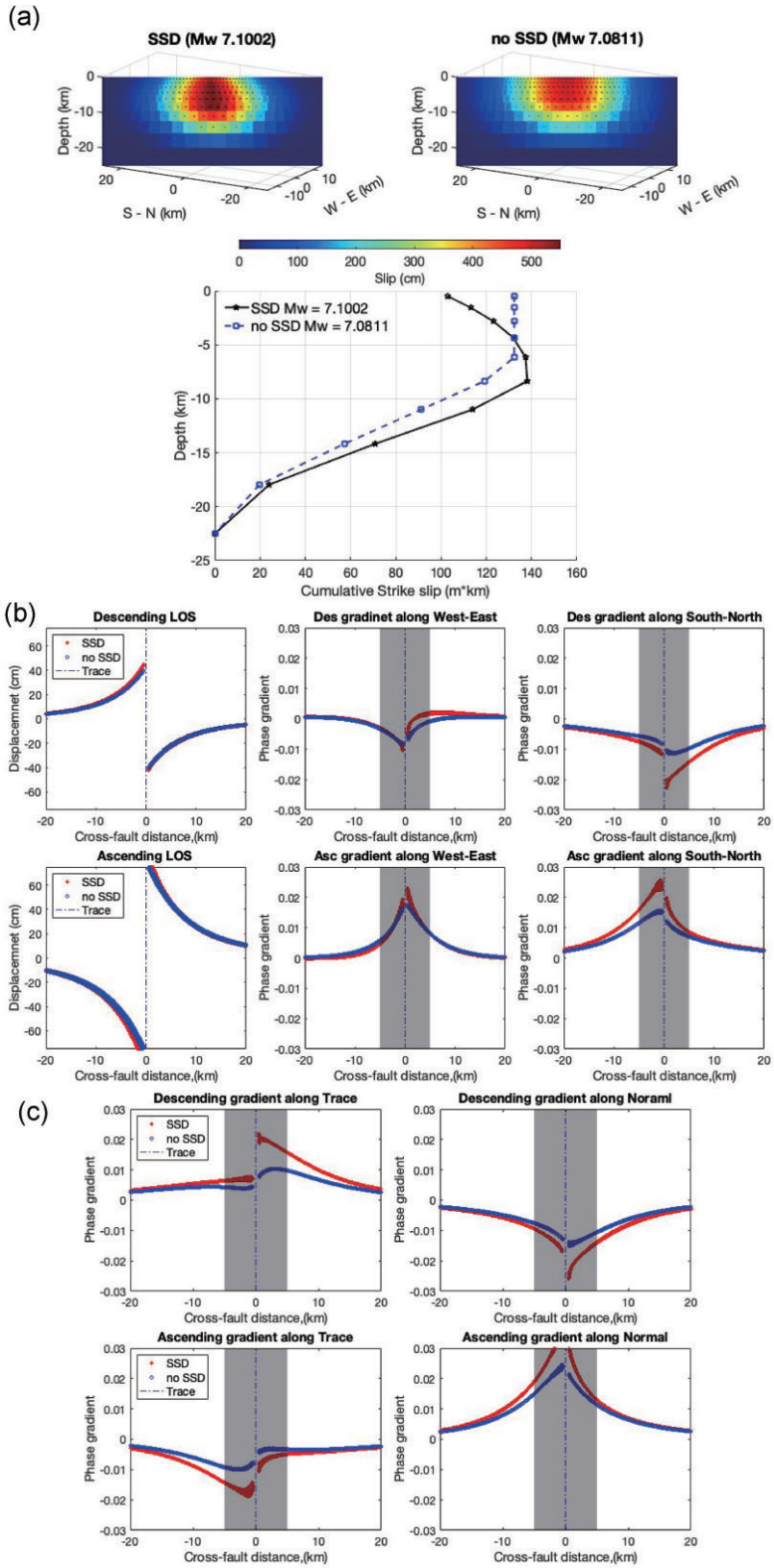

Figure 1. Synthetic models and forwarded LOS displacements and phase gradient profiles along A–A′ (as in Figs S1 and S2, Supplementary Material) comparison from SSD and no SSD model with matched moment. (a) SSD model with 25 per cent SSD, no SSD model, as well as their slip distribution versus depth. (b) LOS displacement profiles A–A′ for the ascending and descending tracks, and the corresponding phase gradient along east to west (x) and south to north (y) directions. Data within 1 km near the rupture are masked. The dash–dotted vertical lines represent location of the fault trace. Blue circles and asterisks show LOS displacement or phase gradient for the corresponding SSD and no SSD models. The grey area is 5-km wide on each side of the fault denoting differentiable phase-gradient pattern when far away from fault. Panel (c) shows the phase gradients projected to the fault trace direction (Trace) and the direction perpendicular to trace (Normal) including ascending and descending tracks.

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radar line-of-sight, LOS deformation gradient) can help construct a slip model with sensitivity to slip at shallow depths. We performed synthetic data analysis to validate the feasibility of incorporating phase gradient data into our kinematic slip modelling and tested its capability of resolving coseismic slip patterns and recovering the true SSD, especially when there is insufficient near-fault data coverage. For real earthquakes, the low signal-to-noise ratio (SNR) of phase gradient data and stronger loss of data coverage from decorrelation usually make the data adoption more complex. However, our application to the 2019 Ridgecrest earthquake sequence shows that it indeed provides a tight constraint on the resulting shallow slip deficit.

2 SYNTHETIC TEST

Here, we construct two synthetic pure strike-slip models in a uniform elastic half-space (Okada 1985) with matched total moment magnitude of 7.1 as shown in (Fig. 1a). Both of them consist of a 60-km-long and 25-km-wide fault plane placed at a 150° strike and 90° dip. The slip along both faults varies as a Gaussian function, tapering to zero towards the ends of the fault trace, but varies with depth such that one produces 25 per cent SSD and one produces no SSD. We separately generated the wrapped phase interferograms with fixed satellite azimuth and look angle for both ascending and descending tracks (Fig. S1, Supplementary Material). Then, we unwrap phase interferograms and calculate the corresponding phase gradient along south to north and west to east directions, project to the fault parallel and normal direction shown in (Figs S1 and S2, Supplementary Material) and separately plot along a profile A–A′—passing through the centre of the fault trace (Figs 1b and c). Our phase gradient profiles show apparent discrepancies, while there are almost indistinguishable phase interferograms for both satellite tracks (Figs 1b and c; Figs S1a, d, g and j, Supplementary Material). Though phase gradient is generally considered to be more sensitive to features with smaller wavelengths, such as densely distributed secondary faults or small surface fractures caused by large ruptures (Xu *et al.* 2020a), the differences in gradient patterns extend to the intermediate scale that is not much influenced by these smaller features, especially after filtering. Here, we incorporate phase gradient information as a new data set to help constrain the slip at shallowest depths for large strike-slip earthquakes, especially in case where near-fault geodetic measurements are absent. The Green's functions calculation with regard to adopting phase gradient contains geographic south–north and west–east direction as follows

$$G_{(\mathrm{grdx})} = \frac{G_{(x2,y)} - G_{(x1,y)}}{\triangle\, x}, G_{(\mathrm{grdy})} = \frac{G_{(x,y2)} - G_{(x,y1)}}{\triangle\, y} \quad (1)$$

where $G_{(x,y)}$ is the Green's function calculated according to analytic function (Okada 1985) on certain surface coordinate corresponding to InSAR data with three components projected into the observation directions. $G_{(\mathrm{grdx})}$ and $G_{(\mathrm{grdy})}$represent the Green's functions for phase gradient along west–east and south–north direction, respectively. Δx, Δy are the spatial intervals for adjacent data points. In case of utilizing gradients generated in radar coordinates, proper projection along radar LOS and the flight direction is needed.

Then we carry out inversions using the exact fault geometry, a non-negative strike-slip constraint and a second-order gradient smoothing (Laplacian) constraint. The data utilized in the inversion are forward calculated LOS displacements added with 5 per cent simulated atmospheric noise of the local signal amplitude and then the phase gradient data are calculated using these displacement grids with 50 m spacing (Do note that smaller spatial interval can accommodate more short-wavelength signals and excessively large spatial interval can lead to an overly smoothed product with lower amplitude, particularly since we apply an additional filtering operation to the product. The impact of different filter scale adopted on inverted SSD will be further discussed in Section 4). In practical scenarios, near-fault gradient information often becomes less reliable due to densely distributed close-to-fault inelastic deformation. As much as this deformation affects radar phase, it leads to an even larger loss of gradient data, since taking gradient essentially magnifies noise. Thus, when running our synthetic test, we forward-calculate LOS displacement and phase gradient separately and introduce three different-size data gaps around the fault, that is, 2-km/6-km/12-km wide gaps (Fig. S3, Supplementary Material). We also experimented with a series of smoothing factors (SF), that is, 0.04, 0.16, 0.64 and phase gradient weights (PGW), that is, 5000, 20 000, 80 000 (Here the large numerical weights are due to the fact that the calculated Green's function of phase gradient is usually small.) to qualitatively test their influences on the inferred SSD.

The results of our inversions to recover synthetic slip distributions with a known SSD are shown in Fig. 2 and Figs S4, S5, S6, and S8, Supplementary Material). In the case of phase data existing up to a 2-km gap (i.e. 1-km gap on each side) close to the fault and low smoothness constraints (i.e. 0.04 and 0.16), the inverted SSD aligns closely with the original SSD due to sufficient near-fault phase constraints even regardless of constraints from the phase gradient data (Figs S4a and b, Supplementary Material). When the model smoothness is set higher (i.e. 0.64), essentially the recovered SSD is less, meaning more slip extends closer to the surface (Fig. S4c, Supplementary Material). Yet by incorporating phase gradient constraints, the actual SSD could be recovered (Figs S4c and S5, Supplementary Material). Do note that different amounts of data masks are applied to the phase gradient (up to 12 km) and the recovery effect remains unaffected (Fig. S5c, Supplementary Material). In the case of a 6- and 12-km wide phase data gap, the SSD with a small SF (i.e. 0.04) given tends to get slightly larger and produces 5–10 per cent artificial SSD (Figs 2a and b) as expected by prior studies (Xu *et al.* 2016; Marchandon *et al.* 2021). But the improvement on recovering true SSD from phase gradient data is slightly more under larger SF (i.e. 0.64) (Fig. S6, Supplementary Material). Essentially, one may get the actual SSD by tuning the smooth factor, yet in reality this parameter is usually chosen in an arbitrary sense. By incorporating phase gradient data, and balancing the weights among the different data set (Fig. S7, Supplementary Material), one can more accurately recover the actual SSD even when a slightly stronger model smoothness is applied (Fig. S6, Supplementary Material).

In order to further investigate and quantify the influence of PGW selection on SSD variation, we examined the model under same data coverage and smoothness and the change of the weight on the gradient data. Similar with the previous analysis, when near-fault phase data are available (e.g. 2-km gap), SSD is firmly constrained even without the inclusion of gradient constraints (Fig. S4b, Supplementary Material). In the case of a larger data gap (6- or 12-km data gap), the role of gradient data becomes more important in constraining SSD, particularly for oversmoothed models (Fig. 2c; Fig. S8c, Supplementary Material). Such recovery effect relies on the choices of PGW and SF (Fig. 2; Fig. S8, Supplementary Material). An excessively high gradient weights and low SF may lead to the emergence of extra SSD artefacts (Fig. 2b; Figs S8a and b, Supplementary Material). Here we quantify the optimal weight selection by balancing the data fitting between phase and phase gradient (Fig. S7, Supplementary Material). In real applications, weights should be determined based on the trade-offs between different data sets

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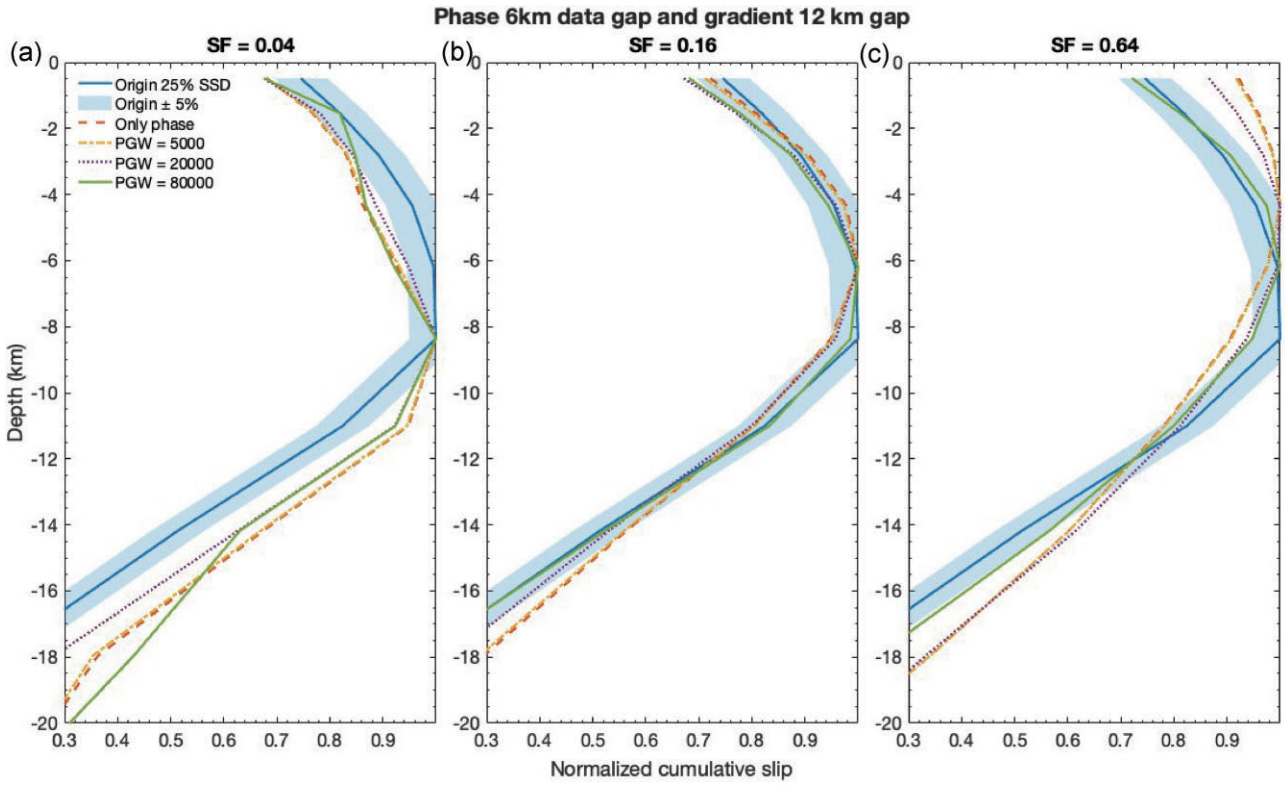

Figure 2. Synthetic tests of SSD variation with changes in PGW under different SF for the case of a 6-km phase data gap and a 12-km gradient data gap. The blue solid line represents the original SSD model, and the light blue shading shows the ±5 per cent range of the original slip distribution versus depth. The yellow dash-dot line, purple dotted and green solid lines indicate SSD variations for PGW values of 5000, 20 000 and 80 000, respectively. Panel (a) shows results for SF of 0.04, panel (b) for SF of 0.16 and panel (c) for SF of 0.64.

by not compromising those data with good quality (Fig. S7, Supplementary Material), which will be discussed in next section.

3 2019 RIDGECREST, CALIFORNIA EARTHQUAKES

The 2019 M_w 6.4 foreshock on July 4 and following M_w 7.1 Ridgecrest earthquake on July 6 occurred in close proximity both spatially and temporally, featuring partially overlapping surface ruptures and aftershock patterns and ruptured a series of complex immature orthogonal fault planes near the Airport Lake and Little Lake fault zones (Ross *et al.* 2019; Ponti *et al.* 2020). The surface rupture and detailed deformation were well imaged by the twin Sentinel-1 satellites. Prior studies have presented a variety of coseismic slip models based on geodetic tools (InSAR, GNSS, optical imagery, etc.) to explore potential rupture mechanism (Ross *et al.* 2019; Fang *et al.* 2020; Jia *et al.* 2020), unknown fault geometry at depth (Feng *et al.* 2020), stress transferring mechanism including the cascading relation between both events (Barnhart *et al.* 2019; Goldberg *et al.* 2020), surface distributed fractures pattern and corresponding formation mechanism (Xu *et al.* 2020b), as well as properties of fault damage zones (Zhou *et al.* 2022; Qiu *et al.* 2023; Xu *et al.* 2023). However, coseismic kinematic slip models for Ridgecrest earthquakes exhibited varying degree of SSD because of variable constraints from near-field observations (Wang *et al.* 2020). Another reasonable explanation is the different smoothing strategies adopted. However, coseismic kinematic slip models for the Ridgecrest earthquakes consistently indicate a reasonable SSD of approximately 20–40 per cent, based on the statistical analysis of multiple published models (Wang *et al.* 2020). These models were constrained by abundant near-fault observations, which are often difficult to obtain in InSAR studies due to decorrelation. Here, we apply the phase gradient information from Sentinel-1 satellites to test whether phase gradient approach can recover the actual SSD of the 2019 Ridgecrest earthquakes without relying on near-field data, similar to the synthetic test.

Our kinematic slip inversion for the earthquake sequence is constrained by a variety of geodetic observations including LOS displacements from Sentinel-1 and ALOS-2 satellites, multi-aperture interferometry (MAI) from ALOS-2 (Xu *et al.* 2020b), GNSS offsets from 92 stations (Klein *et al.* 2019), as well as phase gradient data from Sentinel-1 (see Data Availability). Instead of using optical-offset data, which provides a very strong constraint on shallow slip and could overshadow the contribution of phase-gradient information, we deliberately excluded these data in order to test the effectiveness of the proposed strategy.

The radar interferograms were processed by open source InSAR processing software GMTSAR (Sandwell *et al.* 2016; Xu *et al.* 2017). To generate phase-gradient maps, the data were processed after unwrapping phase and geocoding. For this, the gradient operation is performed to match the definition of Green's function (eq. 1)

$$\nabla\phi_x = \frac{\phi_{x2} - \phi_{x1}}{\nabla x}, \ \nabla\phi_y = \frac{\phi_{y2} - \phi_{y1}}{\nabla y}, \tag{2}$$

where x and y denote the direction of gradient west–east and south–north. First, data from Sentinel-1 ascending T64 and descending T71 were geocoded and sampled with 50 m resolution. Then, we cropped the whole interferogram to a small area covering dominant

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coseimic signal (Fig. S9, Supplementary Material) and calculated the phase gradient separately along south–north and west–east direction using Generic Mapping Tools (Wessel *et al.* 2013). Then, we employed a 400-m Gaussian filter on the original phase gradient and calculated the difference between the original gradient information and the filtered results by setting a threshold to mask those points with significant deviations after filtering in the original gradient observation. This process was repeated to remove most of high-frequency signals corresponding to dense microfractures, inelastic deformation and other non-coseismic signal, which may greatly disrupt the coseismic phase gradient pattern. Subsequently, a 3-km filtering window was applied to smooth the residual high-frequency signals and atmospheric noise etc. Then we used a rectangular mask strategy about 4 km wide for the mainshock fault and 2-km wide for the foreshock fault due to lower amplitude and decorrelations (Fig. S9, Supplementary Material). A quadtree-based downsampling method (Simons *et al.* 2002) was applied to both phase and phase-gradient data. The final subsampled products are shown in the (Fig. S10, Supplementary Material). Before conducting a series of inversion tests, we applied different masking strategies for phase and MAI measurements including progressively reducing data gaps for the mainshock (4, 3, 2 km and no additional mask) and foreshock fault (2 or 1 km, no additional mask). Note the maximum gap corresponds to the effective phase-gradient coverage (Fig. S9, Supplememtary Material). This is to explore how the changes in data coverage of phase gradient data and other geodetic data set affect the SSD. We also considered changing other inversion parameters including SF, and PGW. The fault geometry structure adopted in our inversion is the same as model provided by (Xu *et al.* 2020b) and data weight set for phase, MAI and GNSS are 1, 0.1, 0.1, respectively (as in Xu *et al.* 2020b). Here, we utilized a regularization curve to describe the influence of SF and PGW on the fitting of other data sets in order to determine an appropriate selection.

The inversion approach adopted here largely follows (Xu *et al.* 2016), in which observations from InSAR phase data, phase gradient, MAI data and GNSS data, can be combined into one inversion. The only difference is that we used GNSS data to separate the two events, which is described by the equation bellow,

$$\begin{bmatrix} W_{\mathrm{P}}G_{\mathrm{P}} & W_{\mathrm{P}}G_{\mathrm{P}} & W_{\mathrm{P}}W_{\mathrm{r}}R_{\mathrm{P}} \\ W_{\mathrm{grd}}G_{\mathrm{grd}} & W_{\mathrm{grd}}G_{\mathrm{grd}} & W_{\mathrm{grd}}W_{\mathrm{r}}R_{\mathrm{grd}} \\ W_{\mathrm{a}}G_{\mathrm{a}} & W_{\mathrm{a}}G_{\mathrm{a}} & W_{\mathrm{a}}W_{\mathrm{r}}R_{\mathrm{a}} \\ W_{\mathrm{g}}G_{\mathrm{g}} & 0 & 0 \\ 0 & W_{\mathrm{g}}G_{\mathrm{g}} & 0 \\ \lambda s & 0 & 0 \\ 0 & \lambda s & 0 \end{bmatrix} \begin{bmatrix} m_1 \\ m_2 \\ m_{\mathrm{r}} \end{bmatrix} = \begin{bmatrix} W_{\mathrm{P}}d_{\mathrm{P}} \\ W_{\mathrm{grd}}d_{\mathrm{grd}} \\ W_{\mathrm{a}}d_{\mathrm{a}} \\ W_{\mathrm{g}}d_{\mathrm{g}} \\ W_{\mathrm{g}}d_{\mathrm{g}} \\ 0 \\ 0 \end{bmatrix}$$

where G_{P}, G_{grd}, G_{a}, G_{g} are Green's functions matrix components for InSAR LOS displacements (p), phase gradient (grd), MAI (a) and GNSS displacements (g); W_{P}, W_{grd}, W_{a}, W_{g} are weights solved to provide the maximum overall fitting to different data sets; W_{r} is an extra weight for ramp estimates for each data set; R_{P}, R_{a} are planar ramp functions for each individual data set from different tracks; d_{P}, d_{grd}, d_{a}, d_{g} are corresponding data vectors, with d_{g1} for the M_{w} 6.4 foreshock and d_{g2} for M_{w} 7.1 mainshock; m_1 and m_2 are solved dislocation slip vectors on the fault plane, and m_{r} are ramp parameters for each individual data set.

Fig. 3 illustrates the relationship between SSD variation and the selection of SF and PGW, using all observations with the same data gaps (4 km for mainshock and 2 km for foreshock). The corresponding inverted slip models are shown in (Fig. S11, Supplementary Material). In the Ridgecrest case, the combination of data gaps and relatively strong smoothing can reduce the recovered SSD to values smaller than the true one, or even eliminate it entirely. This effect is particularly evident in our test, since we deliberately removed any on-fault constraints from optical imagery. Consistent with results from our synthetic tests, the incorporation of gradient data can effectively recover the SSD on an excessively smoothed model (Figs 3a–c). Similar as discussed in our synthetic test, in absence of near-fault observations and with excessively high weight to the gradient, an artificial SSD could be induced (Figs 3a–c) and peak slip at depths may get overestimated (Figs S11g, j, k and l, Supplementary Material) probably due to the overfitting of the potential noise. What's different from our synthetic models is that trade-offs among data sets mainly happens between GNSS data and phase gradient data, rather than with phase or MAI measurements (Figs 3d–i). Such differences are likely because phase-gradient is still sensitive to slip away from fault whose coverage area colocates with some GNSS stations. Thus, we suggest the selection of optimal weight for gradient being near the intersection of GNSS and phase gradient fitting (Figs 3g–i) in order to prevent the overfitting of phase gradient with lower SNR. Since our phase gradient is derived from unwrapped phase, selecting an optimal PGW seems difficult by the trade-offs between phase data and phase gradient. Additionally, the fitting of MAI appears largely unaffected by the phase gradient (Figs 3g–i). Once you set a larger PGW and without obvious sacrificing the fitting of other data sets, the next part would be to select the correct smoothness in order to suppress the magnified noise.

Essentially, one could argue that SSD is, to a large extent, controlled by the smoothness applied for the inversion, as with increase of smoothness, SSD will get smaller or even go to zero. Note that the upper limit of smoothness discussed here is still within the range of fitting reasonably well for all data sets (Figs 3d–f). Thus, it is unlikely to determine the uniquely optimal smoothness factor, since this parameter is usually chosen arbitrarily based on statistics mainly due to lack of well-estimated errors on InSAR data (Lohman & Simons 2005). However, the importance of incorporating phase gradient data here is that with the increase of smoothness, the gradient data responds more quickly in its fitting compared with other radar data (Figs 3d–f). One also need to note that the effective smoothness for the model is altered when data weights (such as PGW shown in Figs 3d–f) are changed, and the optimal weight for each data set is affected by the choice of smoothness as well (Figs 3g–i). Therefore, these parameters need to be selected in an iterative way, that is, first a larger PGW and a larger smoothness, then check the PGW to not sacrifice other data fitting and check smoothness to not affect overall data fitting, and so on.

In most cases, obtaining effective gradient data near the fault is more challenging than phase observation which leads the higher loss of near-fault gradient data. Therefore, we progressively incorporate additional near-fault phase and MAI observations, with each increment moving 1 km closer to the fault and without any data gap compared to the previous data set. The resulting SSD variations are shown in Figs S12, S14 and S16(a)–(c) in Supplementary Material, respectively, and the corresponding sets of inverted slip models with different SF and PGW shown in Figs S13, S15 and S17 in Supplementary Material, respectively. The data masked are 3, 2 km and no additional mask for mainshock fault, 2, 2, 1 km and no additional mask for foreshock fault data sets, respectively. The curves depicting the impact of varying SF and PGW on fitting of other data sets are shown in Figs S12, S14 and S16 (d–l) in Supplementary Material. Overall, as expected from our synthetic test, one either needs close to fault data coverage (Figs S16a–c, Supplementary Material) or by incorporating gradient data to help to well constrain the shallow slip (Figs 3a–c). We conducted a sensitivity test of the

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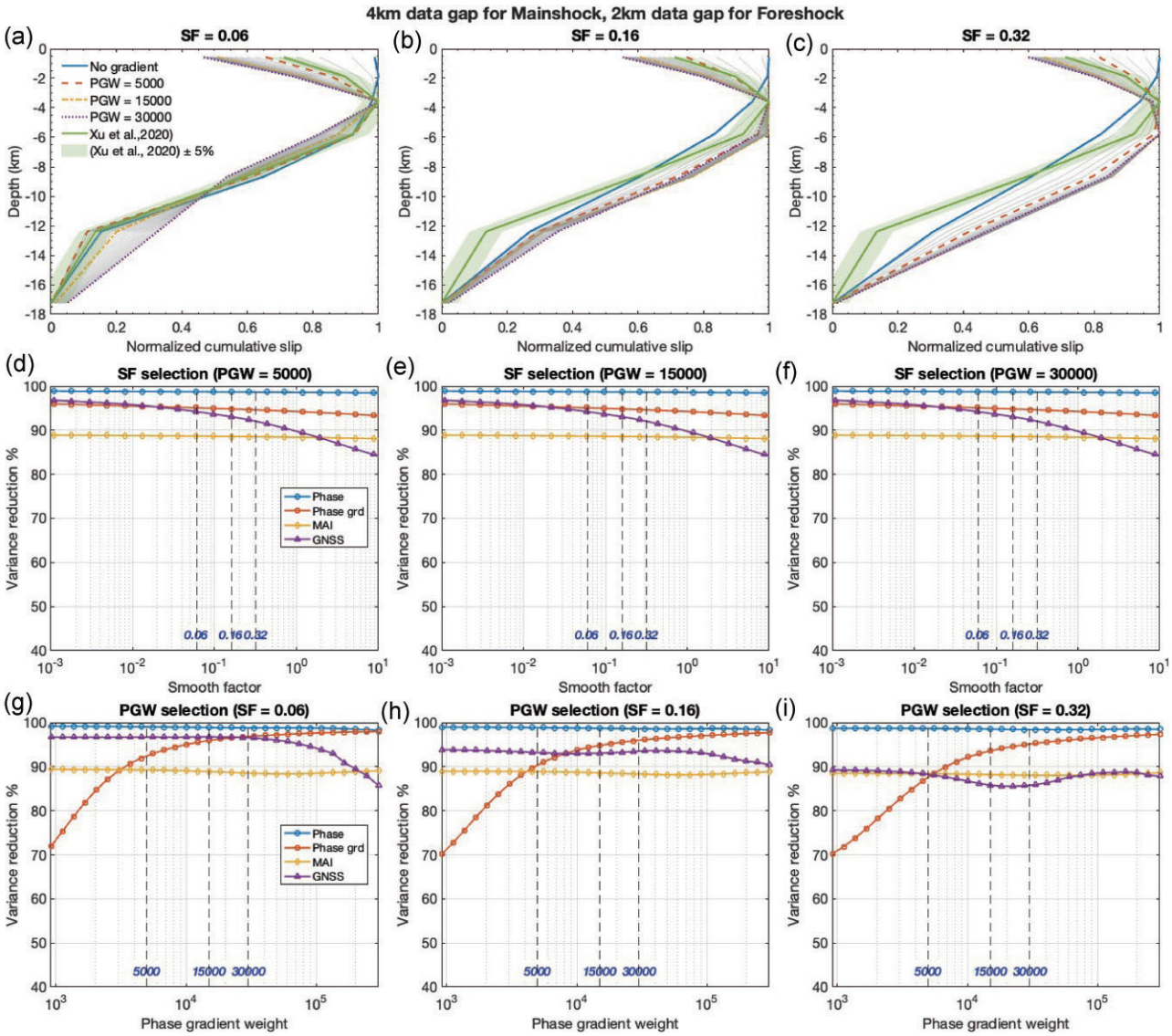

Figure 3. SSD variation and the fitting extent of data sets under different inversion parameter configurations including SF, PGW for the case of a 4-km near-fault mask on the mainshock and a 2-km mask on the foreshock. (a–c) are SSD variations with the change of PGW assigned under the selection of SF 0.06, 0.16 and 0.32, respectively. Highlighted curves indicate models without phase-gradient constraints (blue solid), and with PGW = 5000 (orange dashed), 15 000 (yellow dash–dot), and 30 000 (purple dotted); additional PGW ranging from 0 to 30 000 with 1000 as increment cases are shown as faint grey lines. The reference SSD curve from (Xu *et al.* 2020b) is overplotted with a ±5 per cent green shaded range. (d–f) are the different data set fitting variation with the change of SF given under different PGW set. The black dash lines from left to right represent the SF value we use in panels (a)–(c). Panels (g)–(i) are the different data set fitting variation with the change of PGW given under different SF set. The black dash lines from left to right represent the PGW value highlighted in panels (a)–(c).

SSD estimates under varying near-fault data gaps, smoothness factors (SF), and phase-gradient weights (PGW ranging from 0–30 000 with 3000 as increment) (Fig. 4). The results show increasing SF reduces the SSD estimates, reflecting the enhancement of shallow slip and underestimated peak slip by stronger smoothing, while the variability across different PGW values primarily manifests as a wider spread between minimum and maximum SSD (PGW of 0 versus PGW of 30 000). Compared with the reference SSD value from (Xu *et al.* 2020b), our results suggest that SSD estimates are most consistent when near-fault data coverage is preserved, whereas the presence of wide gaps may artificially inflate the SSD strongly rely on the optimal PGW selection (Fig. 4). In summary, when adopting phase-gradient data, the slip versus depth curve seems to be largely controlled by smoothness (or data weights) which could be set by proposed strategy above. To assess whether the inferred shallow slip is truly constrained by phase gradients rather than inversion artefacts, we conducted model resolution tests and make comparison of surface slips between representative slip models and main geological surface fault right-lateral ruptures documented in (Ponti *et al.* 2020) (Fig. 6a; Fig. S9, Supplementary Material). These tests were performed for (1) models constrained by all available geodetic data sets excluding phase gradients and with near-fault data gaps (i.e. 4 km for the mainshock and 2 km for the foreshock), (2) models using the same data sets but including phase gradients and (3) the prior model from (Xu *et al.* 2020b) (Fig. 5a). Their corresponding SSDs are 0, 33.4 and 29 per cent, respectively (Fig. 6b). Our model resolution tests indicate that the inclusion of phase gradient data enhances the overall resolution of the slip model, with the

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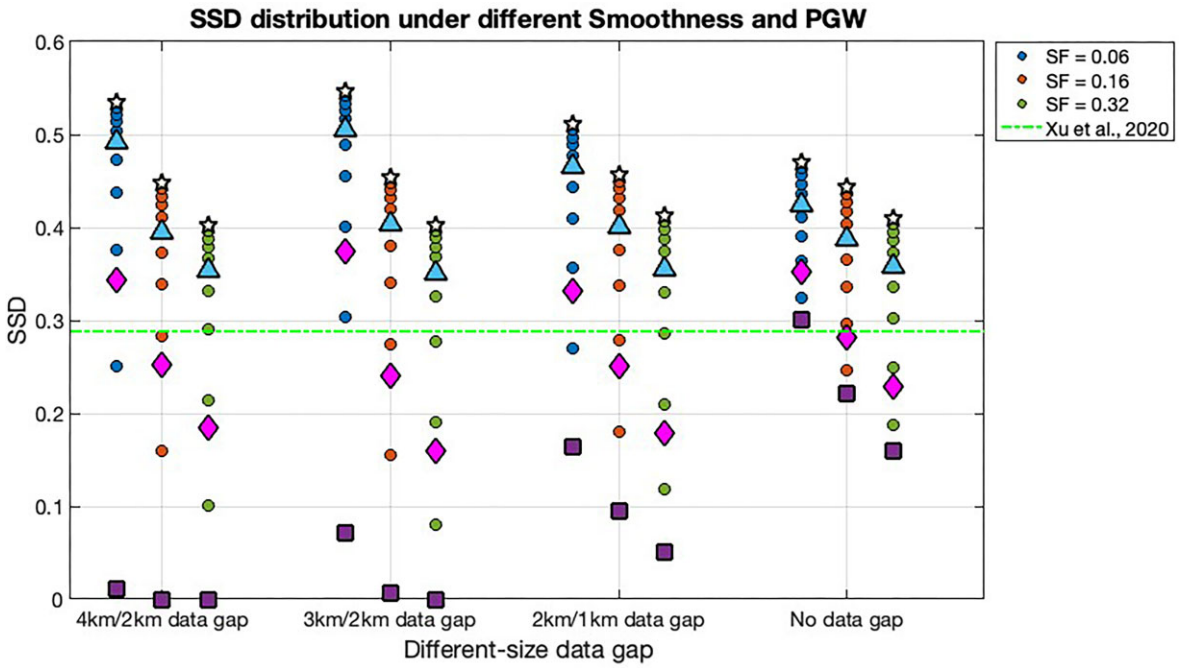

Figure 4. Statistical distributions of SSD values are shown for various data gap configurations, SF and PGW settings. Each vertical cluster corresponds to a specific data gap scenario (4 km for the mainshock and 2 km for the foreshock; 3 and 2 km; 2 and 1 km; and no data gap). Within each cluster, coloured circles represent results for different smoothing factors (SF = 0.06, 0.16 and 0.32) across a range of PGW values (0–30 000, in increments of 3000). Four representative PGW cases from bottom to top are highlighted using enlarged markers with distinct symbols: PGW = 0 (no phase gradient, square), PGW = 5000 (diamond), PGW = 15 000 (triangle) and PGW = 30 000 (pentagram). The green dashed–dotted line indicates the reference SSD value estimated by Xu *et al.* (2020b).

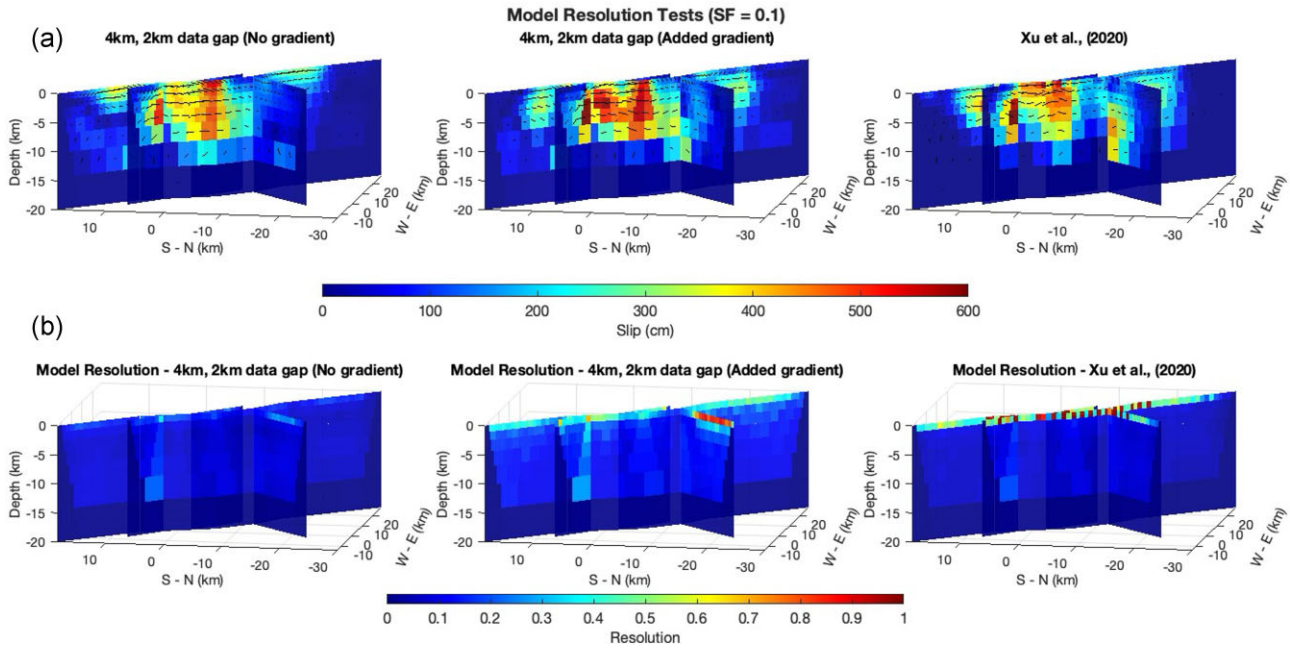

Figure 5. Resolution tests for three slip models (adopted a SF of 0.1): (1) a model constrained by all available geodetic data sets but with near-fault data gaps and excluding phase gradients (4 km for the mainshock and 2 km for the foreshock), (2) the same data set configuration but including phase gradients and (3) the prior model constrained by on-fault optical offset from (Xu *et al.* 2020b). (a) Coseismic slip models for the three cases. (b) Diagonal elements of the corresponding resolution matrices projected onto the fault plane.

improvement being particularly evident in the shallow portion of the fault (Fig. 5b). Without enough near-fault constraints and under the influence of smoothness adopted, the shallow slips are overestimated compared with geological observations (Fig. 6c). However, the surface slips are well constrained, by including phase gradient in inversion even with 4-km near-fault data gap, and almost consistent with geological observations and the model constrained by on-fault optical offset (Xu *et al.* 2020b) (Fig. 6c). Some of the discrepancies between the slip estimated from the slip model constrained by phase gradient with near-fault gaps and the geological observations are likely located in fault barriers or at junctions of complex fault geometries (Figs 6a and c). These deviations may arise from

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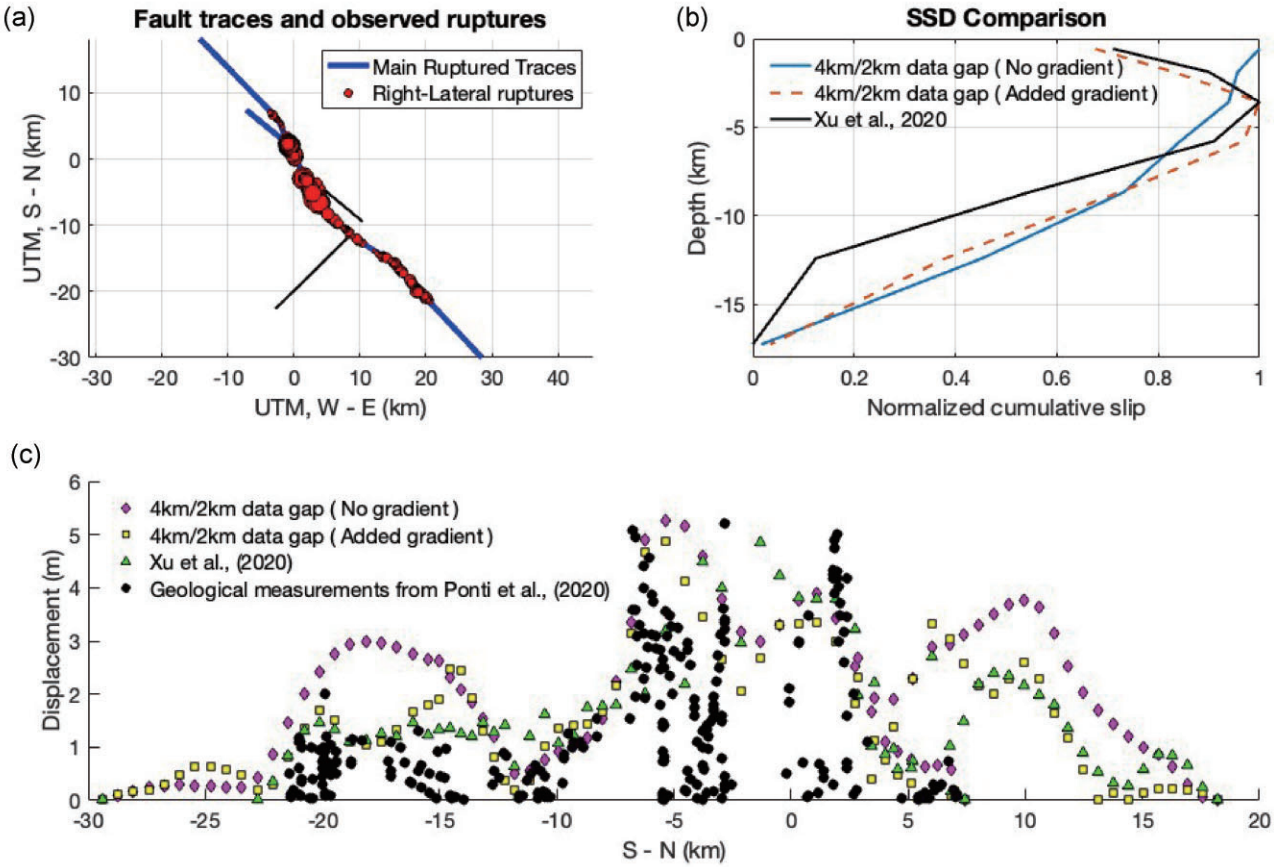

Figure 6. (a) Map of fault traces used in the inversion together with geological right-lateral rupture measurements from (Ponti *et al.* 2020). The red filled circles show locations of field-measured right-lateral offsets and their marker size is scaled to the absolute offset magnitude. Blue solid line segments represent the main ruptured traces. (b) SSD comparison for threes comparative models: (1) a model constrained by all available geodetic data sets but with near-fault masking (4 km for the mainshock and 2 km for the foreshock) and excluding phase gradients, (2) the same data set configuration but including phase gradients and (3) the prior model constrained primarily by optical offsets from Xu *et al.* (2020b). (c) Comparison of surface slip between three slip models (Fig. 5a) and geological measurements documented from Ponti *et al.* (2020) shown in panel (a). Each coloured marker shows the surface slip from one of three models: magenta diamonds—4 km/2 km gap, no phase gradients; yellow squares—4 km/2 km gap with phase gradients included; green triangles—model from Xu *et al.* (2020b). Black filled circles are the geological measurements from Ponti *et al.* (2020).

the loss of information near the fault, or the overfiltering of the phase gradient products, which make it hard to capture localized slip features.

Here, we propose our final coseismic slip model with all data sets except on-fault optical constraints. Here, the model reveals an SSD magnitude of 35 per cent (Figs 7a and b) which is close to prior published models with near-fault data (Wang *et al.* 2020; Antoine *et al.* 2024). This demonstrates the effectiveness of the proposed approach. The trade-offs of smoothness selection (0.1) and phase-gradient weight (8000) are shown in Figs 7(c) and (d) and data residuals in Fig. S18 (Supplementary Material). We compare our preferred model with and without near-fault observation constraints (i.e. 4 km for mainshock, 2 km for foreshock) with same SF, PGW set as well as model with on faults offset constraints from (Xu *et al.* 2020b). We found the corresponding SSDs are 35, 33.4 and 29 per cent, respectively (Fig. 7b). Without enough near-fault constraints and same SF set, the shallow slip could not be strongly constrained, so the smoothing constraint leads to a smaller SSD. However, with the inclusion of phase gradient, the inverted SSD is almost consistent with prior study with on-fault constraints (Fig. 7b). We also invert above-mentioned slip models and the corresponding SSDs by splitting GNSS data for the foreshock and mainshock, respectively, incorporating all available observations, applying the same inversion parameters and compare the corresponding SSDs with the models with data gap and the models from (Xu *et al.* 2020b). We find that the SSD for the mainshock with magnitude of 7.1 are approximately 34, 36.5, 28 per cent and foreshock are approximately 66.2, 34.3 and 27.5 per cent, respectively (Fig. 8). The data residuals are shown in (Fig. S19, Supplementary Material). For the mainshock, all the models have almost consistent slip variation versus depth even without near-fault observations (Fig. 8c). But for the foreshock models, the shallowest slip and seismogenic slip shows apparent difference (Fig. 8d). Such difference will be further discussed next.

4 DISSCUSSION

As pointed out in Section 2, taking gradient essentially magnifies noise. One would normally utilize single best pairs of interferograms for slip inversions (i.e. smallest baseline, shortest time-span, etc.), but to adopt phase-gradient data, it could be more complex. On the one hand, one may have to stack a few pairs of interferograms to reduce the noise level for phase-gradient analysis (Xu *et al.* 2020a), though stacking is essentially easier to do for phase-gradient than radar phase (Sandwell & Price 1998). On the other

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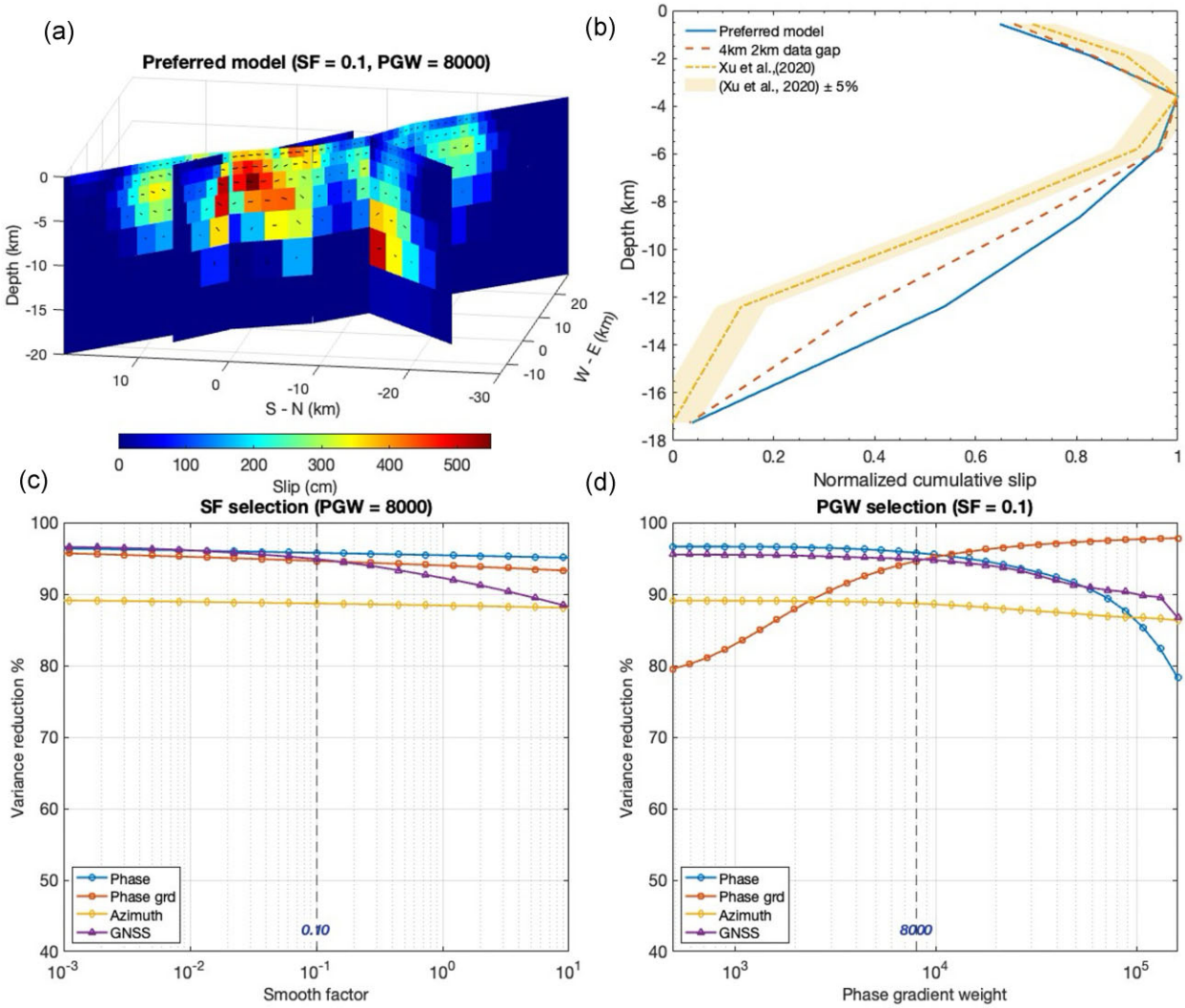

Figure 7. (a) Preferred slip model derived from our available data set under a SF of 0.1 and PGW of 8000. The black short line on each patch marks the corresponding slip direction. (b) SSD comparisons for our preferred model and model with near-fault data gap but including phase gradient as well as optimal model from (Xu *et al.* 2020b), showing roughly 35, 33.4 and 29 per cent SSD, respectively. (c) Variance reduction for data set with the change of SF under the PGW selection of 8000. The black dash lines represent the SF value we use. (d) Variance reduction for each data set versus PGW under the case of SF 0.1 selected. The black dash lines represent the PGW value we use for preferred model (The PGW of 8000 is the intersection between GNSS and phase gradient fitting curve).

hand, phase-gradient is sensitive to decorrelation, and by involving multiple interferograms for stacking, the data coverage may get further reduced. Apart from that, due to the existence of high-gradient, small-scale features in the map, such as deformation on nearby faults (Xu *et al.* 2020a), triggered landslides (Liu & Wang 2023), subsidence due to mining (Tian *et al.* 2024), or any other high-gradient activities, even with good quality, the phase-gradient values may not be actually reflecting our targeted fault slip. One may wonder whether these singular larger measurements will actually affect the slip inversion. Since we are using regularized least-squares to solve for an optimal solution, larger singular misfits get weighted more by the nature of this method (Wang *et al.* 2025). Therefore, it is necessary to remove these records from the phase-gradient before getting it into slip inversions. One could also filter the gradient to a larger wavelength, for example, 2 km, since what constrains fault slip away from fault are the intermediate scale phase-gradient patterns.

The larger shallow deficit for the Ridgecrest mainshock (35 per cent) compared to the 1992 Landers and 1999 Hector Mine earthquakes from nearby Eastern California Shear Zone (Fialko *et al.* 2001; Fialko 2004; Xu *et al.* 2016; Wang *et al.* 2020) could be due to this earthquake sequence happening on a relatively immature fault (Dolan & Haravitch 2014). The complex fractures on nearby faults captured by Sentinel-1 satellites (Xu *et al.* 2020a) imply distributed and inelastic shallow strain release from compliant faults near Ridgecrest that may lead to the larger coseismic SSD (Xu *et al.* 2023). The shallow deficit derived for the foreshock with all available observations is about 66.2 per cent, which when treated as a major rupture, can hardly be compensated by any known mechanisms (Xu *et al.* 2016). Despite the existence of uncertainties in the kinematic inversion, similar or even larger SSD is reported for other intermediate sized strike-slip events, such as the 2004 Bam earthquake (Fialko *et al.* 2005), even though for some of these cases, the data coverage is almost complete. When putting these cases together with a gallery of other larger strike-slip

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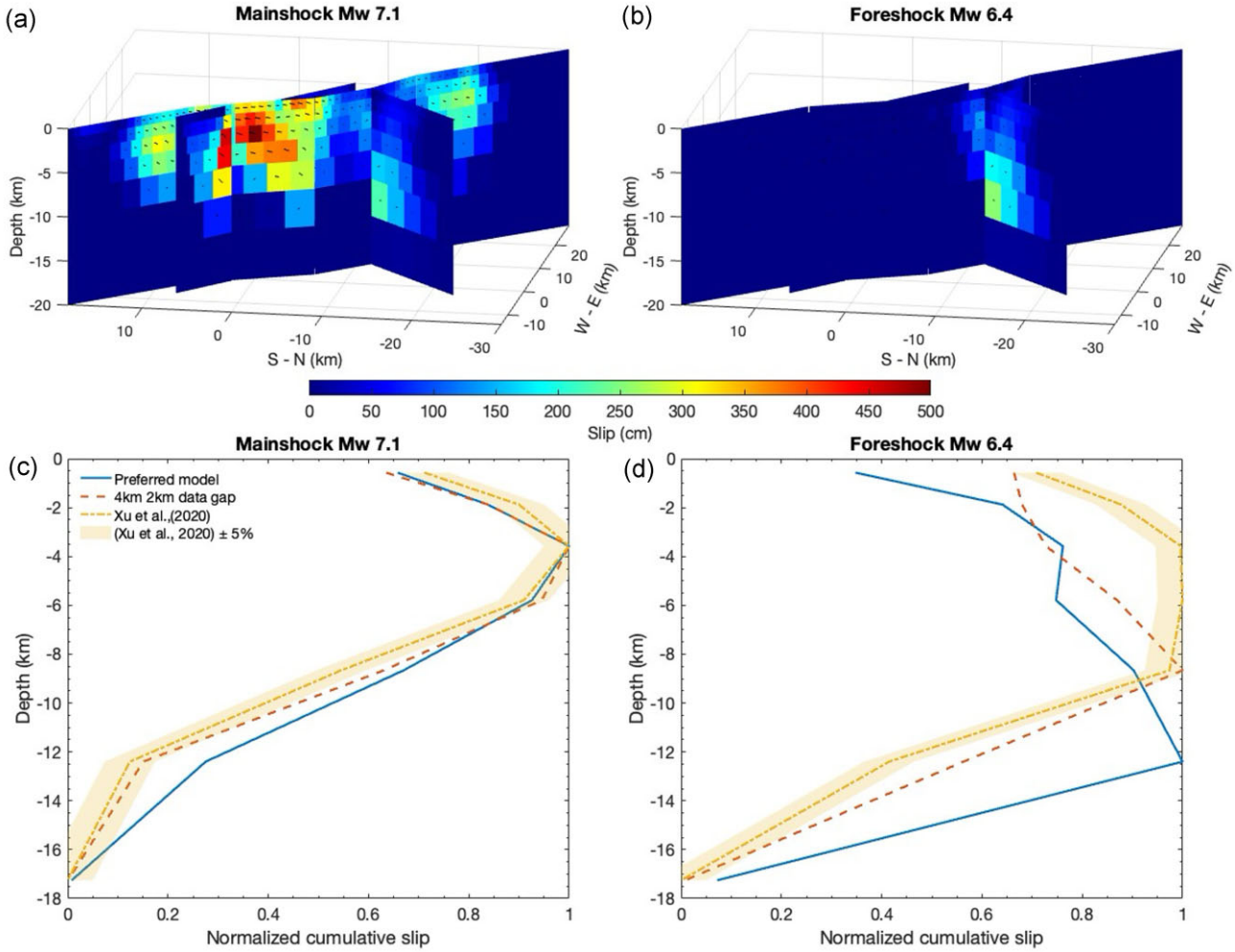

Figure 8. Preferred inverted slip models by splitting the mainshock and foreshock with GNSS data. Panel (a) is preferred M_w 7.1 main shock slip model with probably 33.9 per cent SSD. Panel (b) is referred M_w 6.4 foreshock slip model. (c) Cumulative slip versus depth from our preferred model and model with near-fault data gap as well as optimal model from (Xu *et al.* 2020b) showing roughly 34, 36.5, 28 per cent SSD, respectively. Panel (d) is Cumulative slip versus depth comparison for foreshock showing roughly 66.2, 34.3, 27.5 per cent SSD, respectively. The achieved fitting for each data set has a variance reduction of 96.3, 92, 89 and 96 per cent for InSAR phase, phase gradient, MAI and GNSS data, respectively. The solved corresponding moment magnitude for each earthquake is M_w 6.98 and M_w 6.50, with a 32 GPa assumed average shear modulus.

events, a decay of SSD with the increase of moment magnitude is documented (Antoine *et al.* 2024). Considering excluding on-faults offset constraints on the shallowest slip and our phase gradients are sensitive to the short-wavelength signals, it is difficult to extract true foreshock gradient signal from coseismic mixture products. Our preferred model and the model with data gaps probably accommodate potential noises as part of contribution to slip, leading to less reliable result for the foreshock (Fig. 8d). However, it has to be noted that the discussion about SSD is only meaningful when it comes to the largest earthquakes a fault system can host, as they are responsible for the major release of the accumulated seismic moment. Essentially, SSD should be 100 per cent as long as the magnitude of the earthquake is small enough, since these smaller events can only partially rupture the entire seismogenic zone. But as noted by many prior analyses (Scholz *et al.* 1969), overall, there is no extra seismic activity at the shallowest crust. Thus, these intermediate sized earthquakes should not be considered with any major contribution in the discussion of the problem of shallow deficit.

Additionally, we reproduced phase gradient along the south–north and west–east direction convenient for Green's function calculation instead of projecting to the gradient directions acquired in radar coordinates, as the latter, though with slight better quality, could be complex to implement, especially considering radar line-of-sight changes with range. One may also consider projecting the measurements into fault-parallel and vertical directions, since gradient information perpendicular to the fault strike is more capable of reflecting depth-dependent slip pattern variations (Fig. 1c; Fig. S2, Supplementary Material). However, in reality, the geometry of complex faults can be hard to find an approximate as fault-parallel and fault-vertical direction such as the 2023 Turkey earthquakes doublet. Given the potential deviation of fault geometry in real cases, such as fault dip, we also discuss the potential influence of dip deviations on inferred SSD under different PGW and a large smoothness (Figs S20–S22, Supplementary Material). Our tests show that variations in fault dip indeed affect the estimated SSD, with the influence becoming particularly pronounced when large phase gradient weights are applied (Figs S20c, S21c, S22c, Supplementary

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Material). However, the large deviation of dip may strongly degrade the data fitting, such that fault dip can be well constrained by abundant geodetic observations (Figs S20–S22 inset, Supplementary Material). Considering the fault dip parameter is a nonlinear parameter in inversion, and that phase gradient data may also be sensitive to it, we recommend that researchers first obtain relatively well-constrained fault geometry before incorporating phase gradient constraints into the inversion. And if we have sufficiently reliable near-fault observations, the necessity of including processed phase gradient data should be re-evaluated, as on-fault offset puts very strong constraints on SSD. However, further consideration is required when using high-resolution optical measurements: it remains debatable whether localized off-fault deformation observed in optical imagery should be treated as on-fault offset for inversion purposes, because the minimum patch size in our fault discretization for large events is typically on the order of 1 km and may not resolve very localized features. Moreover, although on-fault offsets are powerful constraints, measurement errors or mis-calculated offsets can produce large biases in the inferred SSD.

Finally, coseismic phase-gradient signals are susceptible to short-wavelength disturbances, which can obscure true deformation gradients so that phase gradient products after different-scale filtering may also influence final inverted SSD. To evaluate that, we performed sensitivity tests on the influence of filtering scale based on our synthetic data (Fig. S23, Supplementary Material). We computed phase gradients in the west–east and south–north directions (Figs S24 and S25, Supplementary Material) and applied Gaussian spatial filters with three kernel scales (400, 800 and 2000 m) to the gradient products. For each filtered product we plotted the A–A′ profiles (Figs S26–S28, Supplementary Material) to inspect how the short-wavelength content is affected by filtering. Next, to assess the impact on inversion results, we ran a suite of slip inversions constrained by the differently filtered gradient data sets. These inversions explored PGW values from 0 to 200 000 with an increment of 5000 under a relatively large smoothness (SF of 0.64) to qualitatively assess the influence of filtering and weighting on the inferred SSD. To mimic realistic near-fault data loss (e.g. decorrelation, surface rupture and high-gradient activities), we masked the near-fault LOS displacement within 6 km and the phase-gradient data within 12 km for each data set (Figs S29–S31, Supplementary Material). The resulting SSD variations for each filter scale are shown in (Figs S32–S34, Supplementary Material). These tests indicate that (i) filtering strength matters principally when the inversion is permitted to fit small-scale structure (i.e. low SF and high PGW) and (ii) appropriate combinations of smoothing and gradient weighting reduce the risk of fitting filtered noise while preserving the gradient signal that constrains shallow slip. Accordingly, we recommend: (1) to test a range of filtering scales when preparing gradient products (this is in fact regularly done in production of phase-gradient data), and (2) avoid overweighting filtered gradients in slip inversions unless the smoothness is sufficiently strong or independent near-fault constraints are available. However, the filtered phase gradient needs to be subsampled before being involved in inversion and the different subsampling strategies (as an equivalent spatial smoothing) could also affect the inferred SSD (Marchandon *et al.* 2021). Thus, how to extract true coseismic gradient signal for area with low SNR and reduce the effect from other high gradient deformation activities requires further investigation.

5 CONCLUSION

We utilize the phase gradient information to jointly invert for coseismic slip, providing an extra geodetic constraint on shallow slip deficit. The relationship between phase gradient and fault slip is mathematically straightforward and can be easily converted with solutions from pure elastic half-space. Our proposed approach shows a strong and effective constraint on SSD both for our synthetic tests with good data quality and for the 2019 Ridgecrest earthquakes with data loss due to near-fault rupture and decorrelation over nearby areas. The acquired SSD for the Ridgecrest earthquakes is comparable to that derived from prior studies, which proves the feasibility of the approach, considering we excluded the help from any near fault constraints. Such phase gradient approach may be applicable to other types of earthquakes.

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SUPPORTING INFORMATION

Supplementary data are available at *GJIRAS* online.

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DATA AVAILABILITY

All SAR acquisitions and subsample data are originally available through the Copernicus Dataspace (https://dataspace.copernicus.eu) and archived at Alaska Satellite Facility (https://search.asf.alaska.edu/), with GNSS data archived at SOPAC (http://sopac-csrc.ucsd.edu). All subsampled data sets and same data sets but with near-fault data gaps, including phase gradient data, are available at (https://doi.org/10.5281/zenodo.17107180).

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